

Session 10: Strings Topics: Declaring and handling strings. Common functions: strlen(), strcpy(), strcmp() . Example: Create a username from name input.

Tasks

- 1) **Declare a string variable called song Title and assign it the value 'Tum Hi Ho'. Print the length of the string using strlen()**

Code-

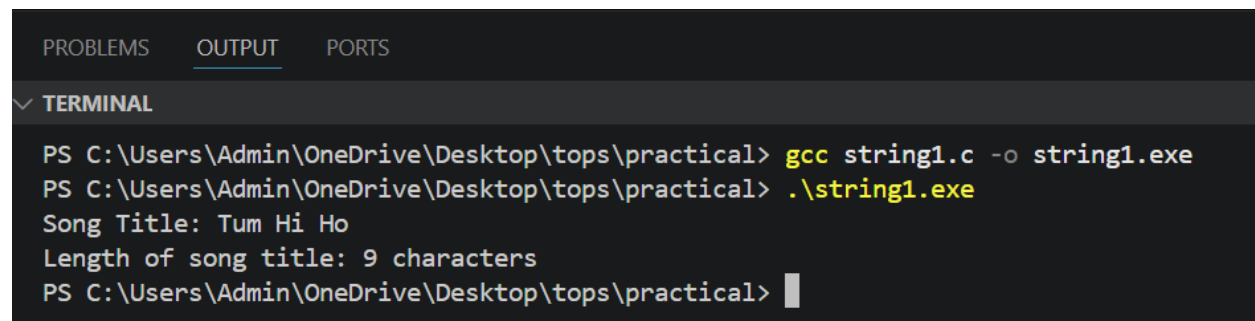
```
#include <stdio.h>
#include <string.h>

int main()
{
    char songTitle[] = "Tum Hi Ho";

    printf("Song Title: %s\n", songTitle);
    printf("Length of song title: %lu characters\n", strlen(songTitle));

    return 0;
}
```

Output-



```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc string1.c -o string1.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\string1.exe
Song Title: Tum Hi Ho
Length of song title: 9 characters
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```

2) Take input for two usernames (as strings) and compare them using strcmp(). Display whether they are the same or different.

Code-

```
#include <stdio.h>
#include <string.h>

int main()
{
    char username1[50];
    char username2[50];

    printf("Enter first username: ");
    scanf("%s", username1);

    printf("Enter second username: ");
    scanf("%s", username2);

    if(strcmp(username1, username2) == 0)
    {
        printf("Both usernames are the same.\n");
    }
    else
    {
        printf("Usernames are different.\n");
    }

    return 0;
}
```

Output:

PROBLEMS OUTPUT PORTS

✓ **TERMINAL**

```
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc stirng2.c -o stirng2.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\stirng2.exe
Enter first username: abhi1234
Enter second username: rahul123
Usernames are different.
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\stirng2.exe
Enter first username: abhi1137
Enter second username: abhi1137
Both usernames are the same.
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> █
```

3) Copy the string 'Flipkart' into another string variable called shopping App using strcpy(), then print the value of shoppingApp.

Hint: Make sure to declare enough space for the destination string.

Code-

```
#include <stdio.h>

#include <string.h>

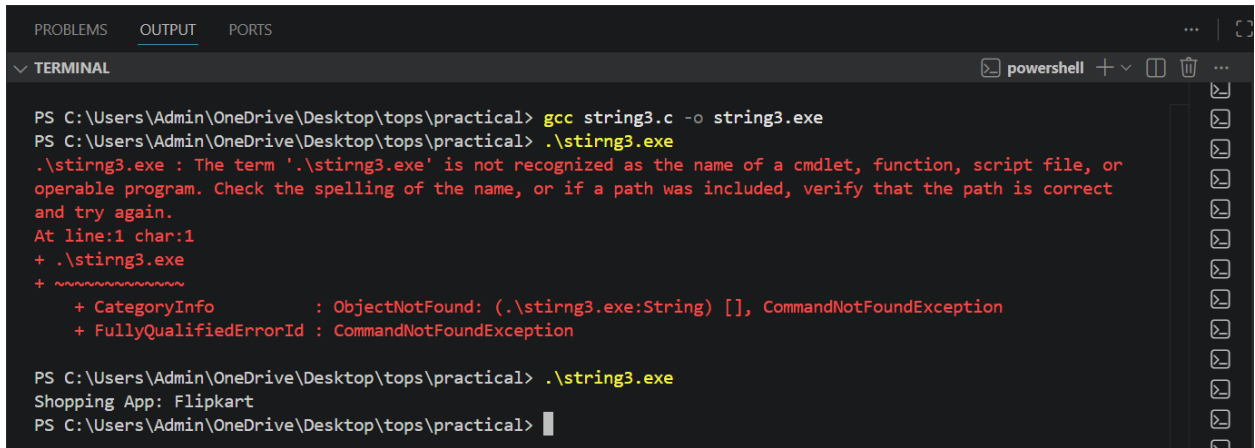
int main()
{
    char source[] = "Flipkart";
    char shoppingApp[20];

    strcpy(shoppingApp, source);

    printf("Shopping App: %s\n", shoppingApp);

    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
TERMINAL  powershell + - ...

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc string3.c -o string3.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\stirng3.exe
.\stirng3.exe : The term '.\stirng3.exe' is not recognized as the name of a cmdlet, function, script file, or
operable program. Check the spelling of the name, or if a path was included, verify that the path is correct
and try again.
At line:1 char:1
+ .\stirng3.exe
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (.\stirng3.exe:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\string3.exe
Shopping App: Flipkart
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> |
```

4) Build a small script that takes the user's full name as input and creates a username by copying only the first 5 characters using `strcpy()`. Print the generated username.

Constraint: If the name is shorter than 5 characters, use the full name as the username.

Code-

```
#include <stdio.h>
#include <string.h>

int main()
{
    char fullName[50];
    char username[10];

    printf("Enter your full name: ");
    scanf("%s", fullName);

    if(strlen(fullName) < 5)
    {
        strcpy(username, fullName);
    }
    else
    {
```

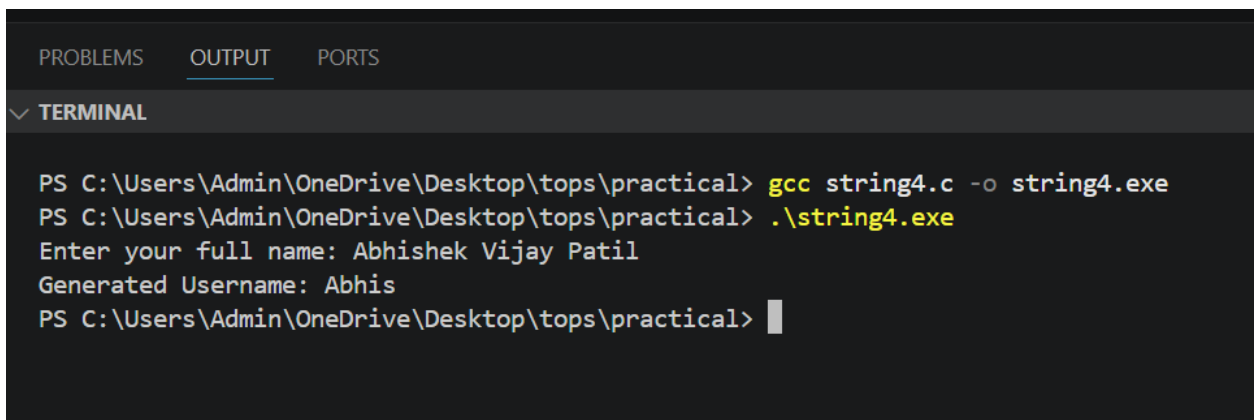
```
    strncpy(username, fullName, 5);

    username[5] = '\0'; // Add null character at end
}

printf("Generated Username: %s\n", username);

return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc string4.c -o string4.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\string4.exe
Enter your full name: Abhishek Vijay Patil
Generated Username: Abhis
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```